

1: Test Connectivity Using Ping

```
Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::d853:f51b:3466:c8a6%6
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:195b:62b3:4949:a318:c28:f064
    Temporary IPv6 Address. . . . . : 2401:4900:195b:62b3:a04d:ad97:417c:6989
    Link-local IPv6 Address . . . . . : fe80::6974:45a0:8874:a935%13
    IPv4 Address. . . . . : 10.183.128.156
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::9409:ff:fe1d:7683%13
                                10.183.128.21

C:\Users\sketan>ping 10.183.128.156

Pinging 10.183.128.156 with 32 bytes of data:
Reply from 10.183.128.156: bytes=32 time<1ms TTL=128
Reply from 10.183.128.156: bytes=32 time<1ms TTL=128
Reply from 10.183.128.156: bytes=32 time<1ms TTL=128
Reply from 10.183.128.156: bytes=32 time<1ms TTL=128

Ping statistics for 10.183.128.156:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2: Run Traceroute

```
C:\Users\sketan>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4829:400::]
over a maximum of 30 hops:

  1     2 ms     2 ms     2 ms  2401:4900:195b:62b3::8f
  2     *         *         *    Request timed out.
  3    148 ms    104 ms    107 ms 2401:4900:8c:2408::2a1
  4    137 ms    101 ms    101 ms 2401:4900:c0:1::5b4
  5     *         *         *    Request timed out.
  6     87 ms    102 ms    200 ms 2401:4900:c0:1::541
  7     *         *         *    Request timed out.
  8    195 ms    101 ms    101 ms 2404:a800:2a00:103::19
  9    119 ms    105 ms    101 ms 2404:a800::167
 10    225 ms    101 ms    101 ms 2001:4860:4829:400::

Trace complete.
```

3: Disconnect One Device and Ping Again

```
C:\Users\sketan>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:
Reply from 10.183.128.156: Destination host unreachable.
Reply from 10.183.128.156: Destination host unreachable.
Reply from 10.183.128.156: Destination host unreachable.
Reply from 10.183.128.156: Destination host unreachable.

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

4: Devices Connected to Home Wi-Fi

```
C:\Users\sketan>arp -a

Interface: 192.168.56.1 --- 0x6
    Internet Address      Physical Address          Type
    192.168.56.255        ff-ff-ff-ff-ff-ff        static
    224.0.0.22            01-00-5e-00-00-16        static
    224.0.0.251           01-00-5e-00-00-fb        static
    224.0.0.252           01-00-5e-00-00-fc        static
    239.255.255.250       01-00-5e-7f-ff-fa        static

Interface: 10.183.128.156 --- 0xd
    Internet Address      Physical Address          Type
    10.183.128.21         96-09-00-1d-76-83        dynamic
    10.183.128.255        ff-ff-ff-ff-ff-ff        static
    224.0.0.22            01-00-5e-00-00-16        static
    224.0.0.251           01-00-5e-00-00-fb        static
    224.0.0.252           01-00-5e-00-00-fc        static
    239.255.255.250       01-00-5e-7f-ff-fa        static
    255.255.255.255       ff-ff-ff-ff-ff-ff        static
```

192.168.56.1 Laptop Address

10.183.128.156 Wifi Address

5: Report

Practical Report: Testing Network Connectivity Using Ping and Traceroute

Objective

The objective of this practical was to connect multiple devices to the same Wi-Fi network and verify communication using the Ping and Traceroute (Tracert) commands. The activity also involved identifying connected devices using the ARP command.

Procedure

1. Connected a Windows 11 laptop and an Android smartphone to the same Wi-Fi router.
2. Found the IP address of the Android phone from the router or network settings.
3. Opened Command Prompt on the laptop.
4. Executed the ping command using the phone's IP address.
5. Verified that all packets were successfully received.
6. Executed the tracert `www.youtube.com` command to observe the route packets followed through the network.
7. Counted the total number of hops to reach the destination server.
8. Disconnected the Android phone from the Wi-Fi network.
9. Executed the ping command again and observed that the packets timed out.
10. Ran the `arp -a` command to list devices connected to the local network and identified each device.

Results

- The laptop successfully communicated with the Android phone while both devices were connected to the same Wi-Fi.
- The Ping command reported 0% packet loss.
- The Traceroute command reached YouTube in approximately 5 hops.
- After disconnecting the phone, Ping failed with the "Request timed out" error.
- The ARP command displayed all devices currently connected to the home network.

